Birth Control by Impairing Fertility:

A Critical Review of Thermal and Intravasal Approaches, with Ethical Boundaries

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Abstract

Approaches to male contraception that operate by impairing fertility—rather than by blocking gamete transport or suppressing the hormonal axis—have been explored for decades. Two strategies have accumulated human data: thermal testicular contraception (TTC), which elevates testicular temperature by approximately 2°C to suppress spermatogenesis, and RISUG (Reversible Inhibition of Sperm Under Guidance), an intravasal polymer injection that damages the sperm membrane. This paper reviews the mechanisms, efficacy, reversibility, and limitations of both approaches. TTC requires 15–24 hours of daily device use, achieves contraceptive thresholds after 2–4 months, and depends on continued use; preliminary data from 34 users show high satisfaction but also transient side effects in nearly half of participants during the first three months. RISUG achieved azoospermia in 82.7% of recipients within two months and in all recipients receiving a complete dose within six months, with no pregnancies reported among 133 complete-dose recipients. Neither method is suited to covert application: both require weeks to months for onset, neither guarantees uniform suppression, and both demand verification through semen analysis. The paper argues that fertility-impairing contraception is scientifically legitimate only within a framework of informed consent, monitoring, and recognition of its current experimental status.

Keywords: male contraception, thermal testicular contraception, RISUG, spermatogenesis, fertility impairment, reproductive coercion, informed consent

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1 Introduction

Male contraceptive options remain limited. Aside from condoms and vasectomy, no method has achieved broad regulatory approval in most jurisdictions, and hormonal male contraception—despite decades of research—remains unavailable for general use. This gap has renewed interest in non-hormonal approaches that target spermatogenesis or sperm function directly.

Two such approaches share a conceptual foundation: they impair fertility by damaging or suppressing sperm production or function, rather than by preventing sperm from reaching the egg through physical barriers or by disrupting the hypothalamic-pituitary-gonadal axis. Thermal testicular contraception (TTC) exploits the temperature sensitivity of spermatogenesis. RISUG exploits the vulnerability of sperm membranes to a polymer deployed within the vas deferens.

This paper examines the evidence for both approaches, evaluates their suitability as contraceptive methods, and addresses a critical distinction: the difference between consensual, monitored fertility impairment and covert application. The latter is neither scientifically reliable nor ethically defensible, and any discussion of fertility-impairing contraception must confront this boundary explicitly.

2 Thermal Testicular Contraception (TTC)

2.1 Mechanism and Method

Spermatogenesis requires a testicular temperature approximately 2–4°C below core body temperature. TTC works by positioning the testes near the external inguinal canal using specialized underwear or a silicone ring, raising testicular temperature by approximately 2°C. This mild hyperthermia induces germ cell apoptosis and suppresses sperm production.

The method was developed in France in the 1980s and has been studied intermittently since. A typical protocol involves wearing the device for 15 to 24 hours per day. One review specifies that daily wear for 15 hours reduces daily sperm production to contraceptive levels within 2 to 4 months, and that this suppression persists only with continued use.

2.2 Efficacy and Reversibility

Human data remain limited. A 2026 mixed-methods study of 34 TTC users reported that 88% achieved the contraceptive threshold—defined as fewer than 1 million sperm per mL—at least once during the study. Earlier work cited in a review of 37 couples using thermal methods as their sole contraception for 6 to 24 months reported no pregnancies, with return to normal sperm production in 3 to 6 months after discontinuation.

However, the evidence base is thin. No device has been approved or standardized, and the 2026 study is explicitly described as preliminary. The contraceptive threshold of <1 million sperm/mL is itself a pragmatic compromise: it does not represent azoospermia, and residual sperm may retain fertilizing capacity in some individuals.

2.3 Tolerability

In the 2026 cohort, 47.1% of participants experienced side effects during the first three months, declining to 6.7% by nine months. These side effects were characterized as acceptable in qualitative interviews, and overall satisfaction exceeded 90%. This suggests that TTC, while demanding in adherence, is tolerable for motivated users.

2.4 Limitations

TTC has three critical limitations. First, onset is slow: months of consistent use are required before contraceptive protection is established. Second, adherence must be continuous; discontinuation leads to gradual return of fertility. Third, effectiveness cannot be assumed without semen analysis, because individual responses vary. These characteristics make TTC unsuitable as an emergency or covert method. A user cannot know whether the method is working without laboratory confirmation, and an unaware partner cannot verify that protection exists.

3 RISUG: Intravasal Fertility Impairment

3.1 Mechanism

RISUG (Reversible Inhibition of Sperm Under Guidance) consists of a styrene maleic anhydride copolymer dissolved in dimethyl sulfoxide, injected into the vas deferens. The polymer is non-sclerosing and non-adherent, remaining in the lumen without occluding it completely. Its contraceptive effect derives from damage to the sperm membrane: the polymer disrupts the acrosome and leaches out enzymes including acrosin and hyaluronidase, rendering sperm incapable of fertilization.

3.2 Efficacy

A phase III clinical study published in 2019 reported outcomes for 139 men, of whom 133 received the complete dose. Among complete-dose recipients, azoospermia was achieved in 22.55% at one month, 82.70% at two months, and 100% by six months. No pregnancies were reported among the 133 complete-dose recipients. Among the six men who received an incomplete dose, two partners conceived, underscoring the dose-dependence of the method.

A separate phase III analysis of 25 subjects found azoospermia in 6 after one month, 15 after two months, 3 after three months, and 1 after four months. Method failure in later phase III reporting has been cited at 2.7%.

3.3 Safety and Reversibility

Reported adverse effects include mild scrotal enlargement, low-grade scrotal pain, and nodule formation at the injection site, all of which resolved within days to weeks. No adverse effects on libido or physical activity were reported.

Reversibility is the method's most significant unresolved question. Reversal has been demonstrated in animal studies using dimethyl sulfoxide and sodium bicarbonate, but reversal of fertilizing ability was not attempted in the 2019 human trial. Without human reversibility data, RISUG functions as a potentially permanent method in practice, despite its name.

4 The Covert-Use Problem

A proposal to use fertility impairment covertly as birth control fails on both scientific and ethical grounds.

Scientifically, neither TTC nor RISUG produces immediate or verifiable contraception. TTC requires months of daily use and periodic semen analysis; a covert user could not confirm efficacy, and the partner could not detect failure until pregnancy occurred. RISUG requires a medical

injection and likewise requires semen testing to confirm azoospermia. An incomplete dose—as occurred in six of 139 participants in the 2019 trial—leaves fertility intact.

Ethically and legally, intentionally impairing a person's fertility without their knowledge constitutes reproductive coercion. Even if the impairment is intended to be temporary, the absence of consent transforms a medical intervention into a harm. The World Health Organization and major reproductive health bodies recognize reproductive autonomy as a fundamental principle; covert fertility impairment violates it directly.

The distinction matters for research and policy. Fertility-impairing contraception deserves study as a consensual option for men and couples who lack alternatives. It does not deserve consideration as a covert tool.

5 Systemic Heat Stress: A Parallel and Asymmetric Pathway

The thermal strategy discussed in Section 2 operates through localized testicular hyperthermia—a targeted, reversible manipulation of a thermally vulnerable anatomical site. A separate body of evidence concerns systemic heat stress (HS): elevation of core body temperature through environmental, occupational, or physiological exposure. The distinction is not merely semantic. Localized heating exploits a male-specific anatomical arrangement; systemic heat stress acts through neuroendocrine, oxidative, and cellular pathways that affect both sexes, but with substantially different evidence bases and risk profiles.

5.1 Mechanisms in the Female Reproductive Axis

A 2025 systematic review of HS effects on the ovary, follicles, and oocytes across mammalian species concluded that heat stress impairs ovarian function through multiple converging mechanisms: disruption of reproductive hormone profiles, reduction in ovarian size and weight, altered ovarian histology, and induction of autophagy. At the cellular level, HS increases granulosa cell apoptosis, impairs folliculogenesis, and reduces oocyte quality and developmental competence.

These findings are mechanistically coherent with a parallel narrative review, which identifies reactive oxygen species (ROS) generation as a central mediator: heat stress overwhelms the antioxidant defenses of the ovarian follicle, producing mitochondrial dysfunction, spindle abnormalities, and DNA fragmentation in the oocyte, with downstream risks of compromised developmental competence and aneuploidy. The same review documents suppression of gonadotropin-releasing hormone pulsatility, leading to altered LH and FSH secretion and consequent anovulation, luteal phase defects, and menstrual irregularities. The pre-ovulatory follicle and corpus luteum are identified as particularly thermosensitive, with reduced steroidogenesis—especially progesterone—and altered endometrial gene expression and uterine blood flow, creating a suboptimal environment for implantation.

The hypothalamic–pituitary–ovarian (HPO) axis is therefore a direct target of systemic HS, distinct from the local spermatogenic suppression that defines thermal male contraception. This mechanistic divergence has a practical implication: while localized testicular heating can be titrated to suppress spermatogenesis without systemic physiological disruption, systemic HS cannot be "dosed" to impair fertility selectively—it engages stress, metabolic, and endocrine pathways simultaneously.

5.2 Epidemiological Signal

Human epidemiological data, while less developed than the animal mechanistic literature, point in a consistent direction. A prospective cohort of 800 pregnant women engaged in moderate-to-heavy occupational work in Tamil Nadu, India, found that women with high occupational heat exposure had a doubled risk of miscarriage (adjusted odds ratio 2.4, 95% CI 1.0–5.7) and elevated risks of any adverse pregnancy or fetal outcome (aOR 2.3, 95% CI 1.4–3.8) and adverse outcome at birth (aOR 2.0, 95% CI 1.2–3.3). A rise in core body temperature was recorded in 17.4% of exposed workers, and 29.6% experienced moderate dehydration, indicating that the exposure was sufficient to produce measurable physiological strain.

Occupational cohorts in greenhouse work and pastry cooking have reported increased menstrual disorders and longer time-to-pregnancy relative to controls. At the population level, higher monthly maximum temperatures have been associated with lower total fertility rates, particularly in regions already above 15–20°C.

5.3 The Male–Female Evidence Asymmetry

The paper's earlier sections established that thermal impairment of male fertility is biologically plausible and experimentally studied as a contraceptive approach. The systemic HS literature reveals a striking asymmetry: the mechanisms of heat-induced male infertility are better characterized than the female equivalent, but the epidemiological signal for adverse pregnancy outcomes is more developed in women. In men, the evidence base supports a contraceptive framing—heat is applied deliberately and its effects on spermatogenesis are the intended outcome. In women, the evidence supports a hazard framing—heat is an involuntary exposure, and its reproductive consequences (miscarriage, fetal growth restriction, birth defects) are harms to be prevented, not effects to be harnessed.

A recent narrative review makes this asymmetry explicit, observing that "current occupational safety guidelines are largely based on male physiology and fail to adequately safeguard female reproductive health." Thermal exposure limits such as WBGT thresholds were developed for a standard 70 kg male performing industrial work and do not account for sex-specific thermoregulation, body size, or reproductive vulnerability. WHO and ILO guidelines acknowledge heat as a general occupational hazard but lack female-specific recommendations for reproductive protection.

5.4 Implications for the Paper's Argument

Incorporating systemic HS into the analysis sharpens three conclusions.

First, it confirms that "fertility impairment by heat" is not a single phenomenon. Localized testicular hyperthermia is a controllable experimental variable; systemic heat stress is an uncontrollable physiological perturbation with reproductive consequences that extend well beyond fertility—into pregnancy maintenance and fetal development. Conflating the two would misrepresent the evidence in both directions: it would overstate the controllability of systemic HS and understate the systemic risks of generalized heating.

Second, it reinforces the paper's rejection of covert fertility impairment. If systemic HS already produces measurable reproductive harm in exposed women—including a doubled miscarriage risk—then any attempt to intentionally impose systemic heat as a covert contraceptive would compound an existing occupational and environmental hazard with deliberate reproductive coercion. The dose–response relationship between heat exposure and adverse pregnancy

outcomes means that "slightly above normal" systemic heating is not a neutral intervention with a narrow effect.

Third, it identifies a policy gap. The absence of female-specific thermal exposure limits is not a technical oversight but a structural one: the standard worker against which safety thresholds are calibrated is male, and the reproductive consequences of heat for women are therefore invisible to the regulatory apparatus. Closing this gap is a matter of reproductive justice, not merely occupational safety.

6 Conclusion

Thermal testicular contraception and RISUG represent two distinct approaches to male contraception by fertility impairment. Both have demonstrated contraceptive efficacy in small human studies. Both require weeks to months for onset, neither guarantees uniform suppression, and both depend on verification that users and partners cannot perform without laboratory testing. TTC demands daily adherence and is reversible; RISUG requires a single injection but lacks human reversibility data.

Systemic heat stress adds a parallel and asymmetric pathway: it impairs female fertility and pregnancy outcomes through neuroendocrine and oxidative mechanisms that cannot be selectively dosed, and it is already a documented occupational and environmental hazard with a doubled miscarriage risk in exposed women.

These methods warrant further research as consensual options. They do not warrant covert application, which is scientifically unreliable and ethically indefensible. The future of fertility-impairing contraception lies in standardization, larger trials, and integration into reproductive care—not in concealment.